

EXPERIMENT 4

AIM: Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)

SOLUTION:

Conversion Functions:

TO_CHAR(): Converts number/date → character.

Query: Convert number to string

```
SQL> SELECT TO_CHAR(123) FROM dual;
```

Output: 123

Query: Convert date to string in format 'DD-MON-YYYY'

```
SQL> SELECT TO_CHAR(SYSDATE, 'DD-MON-YYYY') FROM dual;
```

Output:12-FEB-2026

TO_NUMBER():

Query: Convert string to number

```
SQL> SELECT TO_NUMBER('75') FROM dual;
```

Output:75

TO_DATE():

Query: Convert string to date

```
SQL> SELECT TO_DATE('12-FEB-2026', 'DD-MON-YYYY') FROM dual;
```

Output:12-02-2026

String Functions:

1. Concatenation(||)

```
SQL> SELECT 'Kits' || ' ' || 'College' FROM dual;
```

Output: Kits College

2. LPAD / RPAD : Add characters to left/right of name

Query: Adds * to **left** to make length 6.

```
SQL> SELECT LPAD('SQL', 6, '*') FROM dual;
```

Output: *SQL**

Query: Adds * to **right** to make length 6.

```
SQL> SELECT RPAD('SQL', 6, '*') FROM dual;
```

Output: SQL***

3. LTRIM / RTRIM

LTRIM() : to remove leading (left-side) spaces or specified characters from a string.

```
SQL> SELECT LTRIM(' ORACLE') FROM dual;
```

Output: ORACLE

RTRIM(): remove trailing (right-side) spaces or specified characters from a string.

```
SQL> SELECT RTRIM('ORACLE ') FROM dual;
```

Output: ORACLE

4. LOWER / UPPER / INITCAP

LOWER → small letters

UPPER → capital letters

INITCAP → first letter capital

Query: Change case

```
SQL> SELECT LOWER('DATABASE'), UPPER('database'), INITCAP('kits college') FROM dual;
```

Output: database, DATABASE, Kits College

5. LENGTH / SUBSTR / INSTR

LENGTH → number of characters

SUBSTR → part of a string

INSTR → position of a character

Query: Find length, substring, position

```
SQL> SELECT LENGTH('ORACLE') FROM dual;
```

Output: 6

```
SQL> SELECT SUBSTR('DATABASE', 1, 4) FROM dual;
```

Output: DATA

```
SQL> SELECT INSTR('DATABASE', 'BASE') FROM dual;
```

Output:5

Date Functions:

1. SYSDATE / NEXT_DAY

SYSDATE → current date

NEXT_DAY → next specified day of the week.

Query: Display the current date

```
SQL> SELECT SYSDATE FROM dual;
```

Output:12-02-26

```
SQL> SELECT NEXT_DAY(SYSDATE, 'MONDAY') FROM dual;
```

Output:23-02-26

2. ADD_MONTHS / LAST_DAY

ADD_MONTHS → add months to date

LAST_DAY → last date of month

Query: Add months to current date / find last date of current month

```
SQL> SELECT ADD_MONTHS(SYSDATE, 2) FROM dual;
```

Output:17-04-2026

```
SQL> SELECT LAST_DAY(SYSDATE) FROM dual;
```

Output:28-02-26

3. MONTHS_BETWEEN / LEAST / GREATEST

MONTHS_BETWEEN → months difference

LEAST → smallest value

GREATEST → largest value

Query: Find months difference / smallest / largest value

```
SQL> SELECT MONTHS_BETWEEN(SYSDATE,  
    TO_DATE('01-01-2025','DD-MM-YYYY'))  
FROM dual;
```

Output:13

```
SQL> SELECT LEAST(  
    TO_DATE('12-FEB-2026','DD-MON-YYYY'),  
    TO_DATE('05-JAN-2026','DD-MON-YYYY'),  
    TO_DATE('20-MAR-2026','DD-MON-YYYY')  
) FROM dual;
```

Output:05-01-26

```
SQL> SELECT GREATEST(  
    TO_DATE('12-FEB-2026','DD-MON-YYYY'),  
    TO_DATE('05-JAN-2026','DD-MON-YYYY'),  
    TO_DATE('20-MAR-2026','DD-MON-YYYY')  
) FROM dual;
```

Output:23-03-26

4. TRUNC / ROUND

TRUNC → truncate date to month/year – first day of the month/year

ROUND → round date to month/year – nearest month/year

Query: Truncate or round date to month/year

SQL> SELECT TRUNC(SYSDATE, 'MM') FROM dual;

Output:01-02-26

SQL> SELECT ROUND(SYSDATE, 'YY') FROM dual;

Output:01-01-26